

HEMALATHA V

Python Developer | Backend Development & AI/ML

Chennai, Tamil Nadu, India | 7548836978 | hema1107latha@gmail.com | linkedin.com/in/hema-2311-latha

PROFESSIONAL SUMMARY

Python Developer with 3+ years of experience building backend systems, REST APIs, and automation tools, with hands-on exposure to machine learning workflows using Pandas, NumPy, and scikit-learn. Delivered 10+ production automation tools that cut manual effort by over 40%, built cross-region sync systems across India, Malaysia, and Singapore, and currently design backend tools and tracking systems for a production environment handling large-scale asset and data workflows. Comfortable across the full SDLC — API design, database modeling, cloud deployment on AWS (EC2, S3), and version control with Git — and actively building applied ML projects in classification and predictive modeling.

TECHNICAL SKILLS

Languages: Python, SQL

Backend Frameworks: Django, FastAPI, REST API design, JSON, Postman

AI / ML & Data: Pandas, NumPy, scikit-learn, Matplotlib, basic TensorFlow/Keras, EDA, model evaluation (accuracy, precision/recall, RMSE)

Databases: MySQL, PostgreSQL, SQLite, DBeaver, query optimization

Cloud & DevOps: AWS (EC2, S3), Git, CI-friendly deployment basics

Automation & Tooling: PyQt5 (GUI), Python scripting for workflow automation, third-party API integration

Core Concepts: OOP, Data Structures, SDLC, Automation, API Integration

PROFESSIONAL EXPERIENCE

Python Developer — Spectre Post Private Ltd

Jun 2026 – Present | Tech Stack: Python, PyQt5, REST API, MySQL

- Build and maintain backend tools and internal APIs that automate rendering, asset handling, and file management workflows across a large production environment.
- Design and maintain tracking/monitoring systems that give teams real-time visibility into task status and progress across distributed workflows.
- Write Python automation scripts that reduce manual, repetitive work and improve turnaround time on recurring processes.
- Build integrations across internal systems to keep data and files consistent and synchronized across departments.
- Collaborate with cross-functional teams to translate operational requirements into reliable backend tools, and own debugging and support with minimal downtime.

Junior Python Developer — Sparky Entertainment Pvt Ltd

Jan 2023 – May 2026 | Tech Stack: Python, Django, FastAPI, REST API, PyQt5, MySQL, Git

- Built and maintained 10+ Python automation tools, cutting manual task time by an estimated 40% and reducing recurring errors across teams.
- Designed user-friendly PyQt5 GUI applications that let non-technical users run internal tools independently, reducing dependency on engineering support tickets.
- Built internal REST APIs in FastAPI/Django to support tool-to-tool communication and structured data exchange across production systems.
- Applied OOP principles to deliver modular, reusable code, reducing duplicate logic across 10+ internal tools.
- Partnered with cross-functional teams to translate requirements into working software, and owned debugging, testing, and deployment end-to-end.

PROJECTS

Customer Churn Prediction (Machine Learning) — Personal Project

Tech Stack: Python, Pandas, scikit-learn, Matplotlib, Jupyter

- Built an end-to-end ML pipeline to predict customer churn from a telecom-style dataset, covering data cleaning, feature engineering, and model training.
- Compared Logistic Regression, Decision Tree, and Random Forest models; selected the best performer using accuracy, precision/recall, and confusion matrix analysis.
- Visualized churn drivers and model performance with Matplotlib to make findings interpretable for non-technical stakeholders.

Support Ticket Text Classifier (NLP) — Personal Project

Tech Stack: Python, scikit-learn, NLTK, FastAPI

- Built a text classification model to auto-tag support tickets by category using TF-IDF features and a scikit-learn classifier, served through a FastAPI endpoint for real-time inference.
- Evaluated model quality with precision, recall, and F1-score across ticket categories.

Multi-Region Tool Sync & Notification System

Tech Stack: Python, REST API, JSON, Microsoft Teams API, MySQL

- Synchronized internal tools across India, Malaysia, and Singapore offices via REST APIs, keeping environments consistent and automating version updates across distributed servers.
- Integrated Microsoft Teams API for real-time sync notifications, cutting manual status check-ins.

Task Management System (FastAPI) — Personal Project

Tech Stack: Python, FastAPI, REST API, PostgreSQL, Postman

- Built a RESTful task management API in FastAPI supporting full CRUD operations — creating, updating, assigning, and deleting tasks — backed by a PostgreSQL database.
- Designed the database schema and query layer to handle task ownership, status, and due dates efficiently as data volume grows.
- Structured all request/response contracts in JSON with clear validation, and tested every endpoint in Postman to confirm correct behavior before integration.

AWS File Upload & Management System — Personal Project

Tech Stack: Python, AWS EC2, AWS S3, REST API, JSON

- Built a backend service to upload, store, retrieve, and delete files using AWS S3 as the storage layer, exposed through a REST API.
- Deployed the application on an AWS EC2 instance to simulate a real production cloud environment end to end.
- Implemented secure file handling (validated uploads, controlled access) to prevent unauthorized file operations.

Multivendor Marketplace Management System — Personal Project

Tech Stack: Python, SQLite, REST API, JSON

- Built a backend system to manage multiple vendors and their product catalogs, supporting full CRUD operations on products.
- Implemented an authentication system for secure vendor login, keeping each vendor's access scoped to their own data.
- Designed the database structure to efficiently handle relationships between vendors, products, and inventory.

EDUCATION

Master of Computer Application (MCA) — Madurai Kamaraj University (Distance) | 2022 – 2024 | 83%

B.Sc. Computer Science — Madurai Kamaraj University (Full Time) | 2019 – 2022 | 81%